Clustering Penguins – Unsupervised Machine Learning Report

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Module: Week 7 – Standardisation & Normalisation

Algorithm Used: K-Means Clustering

1. Introduction

This project applies K-Means clustering to the Palmer Penguins dataset using numeric morphological features. The objective is to explore unsupervised learning and evaluate the effect of feature scaling techniques such as standardisation and normalisation.

2. Dataset and Features

- bill_length_mm (culmen_length_mm)
- bill_depth_mm (culmen_depth_mm)
- flipper_length_mm
- body_mass_g

The species column was not used for training, as clustering is an unsupervised learning technique. It was used only for evaluation purposes using the Adjusted Rand Index (ARI).

3. Methodology

- Baseline K-Means clustering without scaling
- Standardisation using StandardScaler (mean=0, std=1)
- Normalisation using MinMaxScaler (range 0–1)
- Dimensionality reduction using PCA for visualisation

4. Results and Analysis

The baseline model showed moderate clustering alignment. However, because K-Means is distance-based, features with larger magnitudes (such as body_mass_g) dominated the clustering outcome.

After applying standardisation, clustering performance improved, demonstrating the importance of feature scaling in distance-based algorithms. Normalisation also improved performance, although results varied slightly depending on feature distributions.

PCA visualisation provided a clear 2D representation of cluster separation and showed that certain species were more distinct in morphological space.

5. Conclusion

This project demonstrates that preprocessing is essential for clustering performance. Standardisation significantly improves the stability and interpretability of K-Means clustering. The experiment reinforces that scaling should be considered a critical step when working with distance-based machine learning algorithms.

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